

TCP/IP Model (Transmission Control Protocol / Internet Protocol)

The **TCP/IP model** is a networking model used to explain how data travels from one device to another over a network such as the **Internet**.

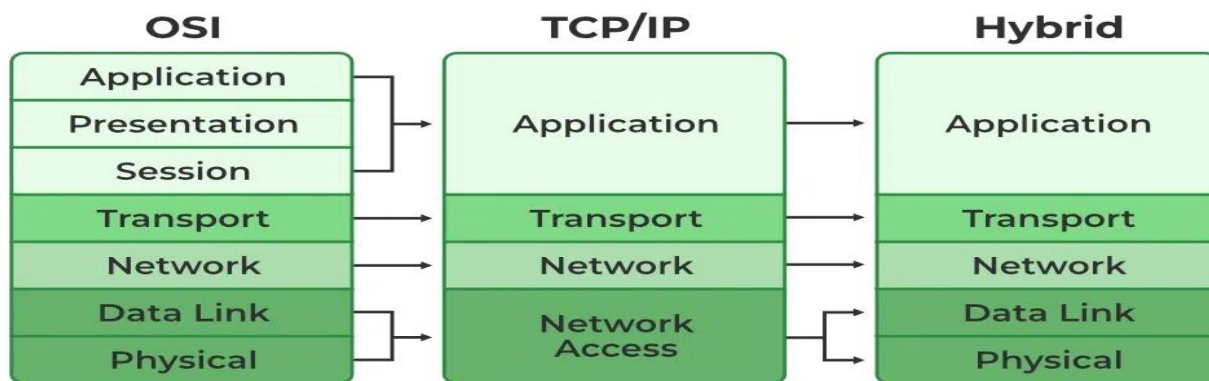
Application Layer: It provides network services directly to applications used by the user.

It is the topmost layer of the TCP/IP Model. This Layer is the combination of the Application, Presentation, and Session Layer of the OSI Model.

Transport Layer: This layer is similar to the Transport layer of the OSI Model. It has the work of Error-Free Message Delivery.

Internet Layer: This layer comprises of Network Layer of the OSI Model. It has the work of moving data packets from destination to source.

Network Access Layer: It is the lowest layer of the TCP/IP Model which comprises of Physical and Data Link Layer of the OSI Model. It helps in data transmission among the same network.



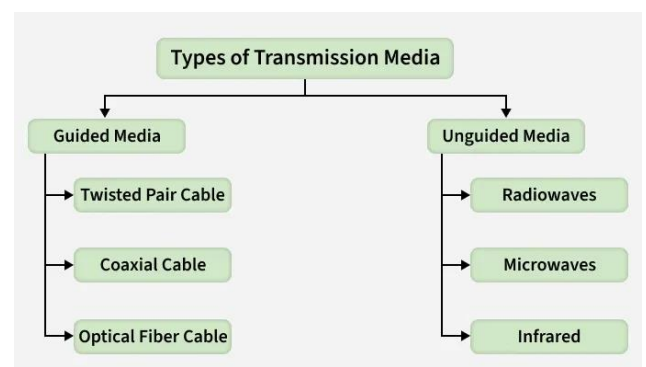
Transmission Media in Computer Networks

Last Updated :5 Jan, 2026

- Transmission media refers to the physical or wireless communication channel used to carry data signals from one device to another within a computer network. It forms the fundamental pathway through which information is transmitted, ensuring connectivity between networked devices.

The selection of a transmission medium depends on factors such as:

- Transmission distance
- Data transfer speed (bandwidth)
- Susceptibility to interference and noise
- Cost and installation requirements



Guided Media

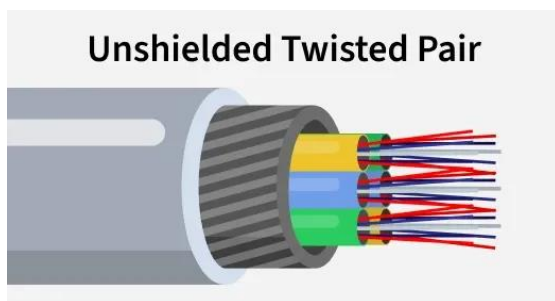
Guided Media also known as wired or bounded transmission media, refers to transmission media in which data signals are transmitted through a physical path using cables. The signal is confined and guided along a fixed route, providing controlled communication between network devices.

- Uses physical cables to transmit data signals.
- Provides dedicated and well-defined transmission paths.
- Offers higher data transmission rates compared to most wireless media.
- Provides better security due to physical connectivity and limited signal leakage.
- Suitable for short to long-distance communication, depending on the cable type used.

1. Twisted Pair Cable :

A **twisted-pair cable** consists of two insulated copper wires twisted around each other.

a. **Unshielded Twisted Pair (UTP):** UTP cable consists of twisted copper wire pairs without any additional metallic shielding. The twisting provides basic protection against interference, making UTP suitable for environments with moderate noise levels.



Advantages:

- Least expensive
- Easy to install and maintain due to its lightweight and flexible nature.
- Supports high data transmission rates over short distances.

Disadvantages:

- Lower performance and capacity compared to

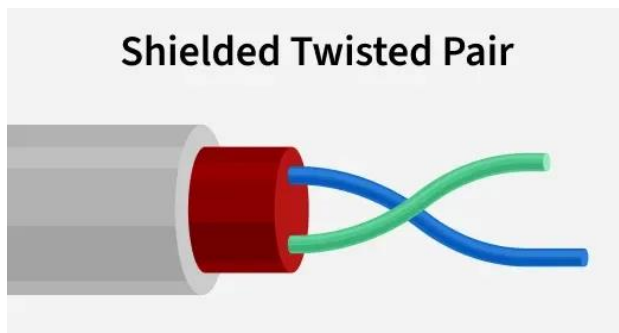
Shielded Twisted Pair (STP).

- Limited transmission distance due to signal attenuation.

(b) **Shielded Twisted Pair (STP):** Shielded Twisted Pair (STP) cable consists of twisted pairs of insulated copper conductors enclosed within an additional metallic shielding layer, such as a foil or braided shield. This shielding provides enhanced protection against external electromagnetic interference

Advantages

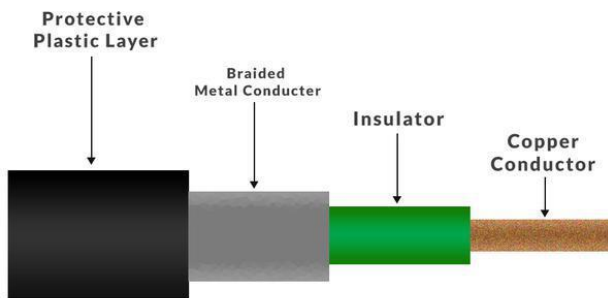
- Provides better performance at higher data transmission rates compared to UTP.
- Significantly reduces electromagnetic interference and crosstalk.
- Ensures more reliable and stable communication, especially in noisy environments.
- Suitable for high-speed Ethernet networks and voice and data transmission systems.



Disadvantages

- Higher cost compared to Unshielded Twisted Pair (UTP).
- More complex installation and termination due to shielding requirements.

2. Coaxial Cable : Coaxial Cable consists of a central copper conductor surrounded by a dielectric insulating layer, a metallic shielding layer, and an outer protective jacket. This structure allows signals to be transmitted with better protection against noise and interference compared to twisted pair cables.



Advantages:

- Supports higher bandwidth
- Offers better signal quality and reliability.
- Less affected by external electromagnetic interference.

Disadvantages:

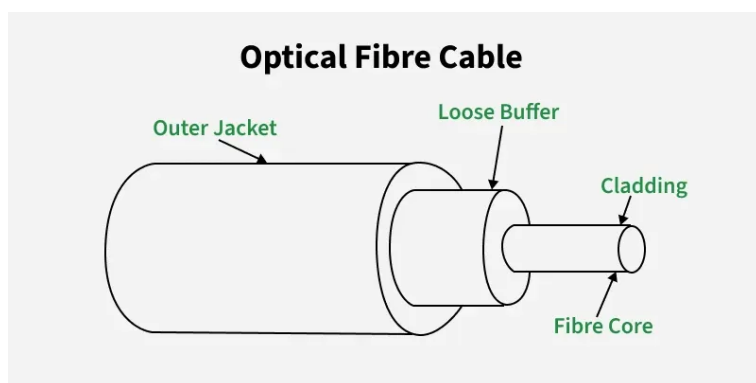
- More expensive
- Requires proper grounding to ensure safety

and minimize interference.

- Bulkier and less flexible due to multiple layers.
- More vulnerable to security breaches, as the cable can be physically tapped.

3. Optical Fiber Cable : Optical Fiber Cable is a guided transmission medium that transmits data in the form of light signals through a glass or plastic core using the principle of total internal reflection. The core is surrounded by a cladding layer with a lower refractive index, which confines the light within the core and enables high-speed, long-distance data transmission.

- Supports very large data volumes at extremely high speeds.
- Can operate in unidirectional or bidirectional communication modes.
- Widely used where high bandwidth, long distance, and low signal loss are required.



Advantages:

- Provides very high bandwidth and data-carrying capacity.
- Lightweight and compact, making installation easier over long distances.
- suitable for long-distance communication.

Disadvantages:

- Complex installation and maintenance due to precise splicing and handling requirements.
- High initial installation and equipment cost.
- Fragile compared to copper cables and requires careful handling.

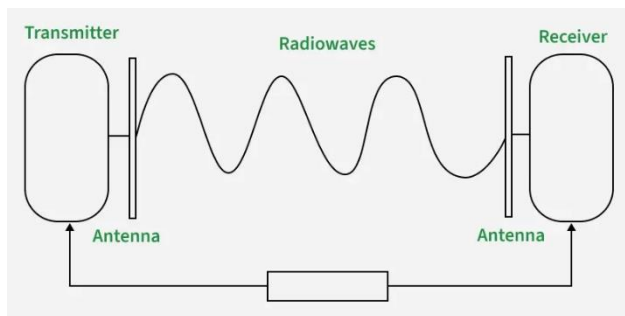
Unguided Media

Unguided media, also known as wireless or [Unbounded transmission media](#), uses electromagnetic waves to transmit data without any physical medium. Signals propagate through free space such as air or vacuum. The main types of unguided media are radio waves, microwaves, and infrared waves.

Features:

- Signals propagate through air or free space
- Less secure due to broadcast nature
- Suitable for long-distance communication

1. Radio Waves : [Radio waves](#) are a type of electromagnetic wave that can easily be generated and can propagate through buildings and other obstacles. They do not require line-of-sight between transmitting and receiving antennas, making them highly suitable for broadcast communication and wireless data transmission.



Frequency Range:
3 kHz – 300 GHz

2. Microwaves

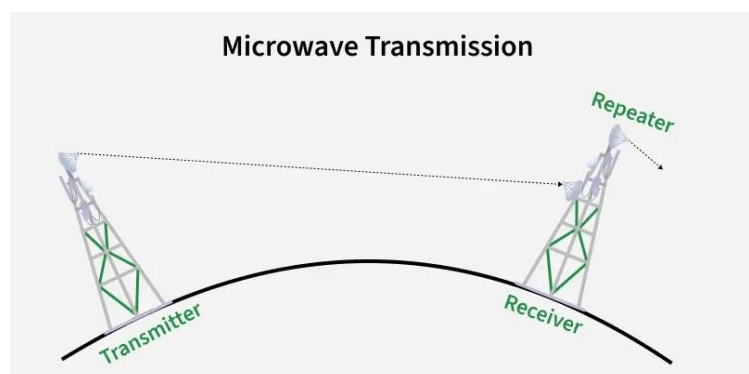
[Micro waves](#) are a form of unguided transmission media that use line-of-sight communication, where the transmitting and receiving antennas must be properly aligned. The transmission range depends on the height of the antennas. Microwaves operate in the frequency range of 1 GHz to 300 GHz and are widely used in mobile communication, satellite links, and television distribution.

Advantages:

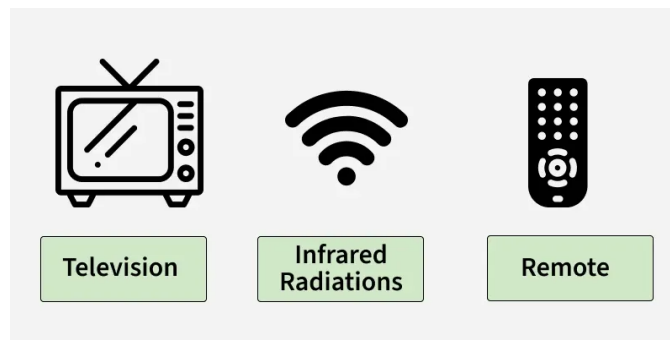
- Cost-effective compared to laying physical cables
- No need for land acquisition
- Suitable for communication over difficult terrains and oceans
- Supports high data transmission rates

Disadvantages:

- Less secure without proper encryption
- Susceptible to weather conditions such as rain and fog
- Affected by obstacles due to line-of-sight requirement
- High cost of antenna design, installation, and maintenance



3. Infrared : Infrared waves are used for short-range wireless communication and operate in the frequency range of 300 GHz to 400 THz. They cannot penetrate solid obstacles, which limits their range but helps minimize interference between nearby systems. Infrared waves are commonly used in TV remote controls, wireless keyboards, mice, and printers.



Narrowband and Broadband Communication Systems

In **Computer Networks**, communication systems can be broadly classified based on the **bandwidth** they provide:

1. **Narrowband communication**
2. **Broadband communication**

1. Narrowband Communication

Narrowband means a communication system that uses a **relatively small range of frequencies (bandwidth)** and generally provides a lower data rate.

Characteristics

- Small bandwidth
- Lower data transmission rate
- Suitable for small amounts of data
- Older communication technology
- Usually higher delay compared with broadband

2. Broadband Communication

Broadband means a communication system that provides a **much larger bandwidth**, allowing data to be transmitted at much higher speeds.

Broadband can support:

- Web browsing
- HD/4K video streaming
- Video calls
- Online gaming
- Large file downloads
- Cloud applications

Feature	Narrowband	Broadband
Bandwidth	Low	High
Data rate	Low	High
Capacity	Low	High
Internet usage	Limited	Extensive
Video streaming	Difficult/limited	Easily supported
Large file transfer	Slow	Fast
Example	Dial-up	Fiber, cable, 4G/5G

In a **connectionless service**, there is **no connection establishment before sending data**.

Each packet is sent independently.

Think of it like **sending separate letters through the postal system**.

Packet 1 →
Packet 2 →
Packet 3 →

The sender doesn't first establish a dedicated connection with the receiver.

Example: UDP

UDP (User Datagram Protocol) is connectionless.

Features

- No connection establishment
- Faster

Design Issues for Layers in Computer Networks

The major design issues that must be considered while designing layers in a computer network are:

Sr. No.	Design Issue	One-Line Explanation
1	Addressing	Identifies the sender, receiver, and specific application using addresses such as MAC address, IP address, and port number.
2	Error Control	Detects and handles errors or lost data during transmission to ensure correct communication.
3	Flow Control	Controls the rate of data transmission so that a fast sender does not overload a slow receiver.
4	Sequencing	Ensures that data packets are received and arranged in the correct order at the destination.
5	Routing	Determines the best path through the network for delivering data from the source to the destination.
6	Multiplexing and Demultiplexing	Allows multiple applications to share a network connection and delivers received data to the correct application.
7	Connection Establishment and Termination	Establishes a communication connection before data transfer and properly terminates it after communication is completed.

Sr. No.	Design Issue	One-Line Explanation
8	Fragmentation and Reassembly	Divides large data packets into smaller pieces for transmission and combines them again at the destination.
9	Data Encapsulation	Adds control information such as headers and trailers to data as it moves through different network layers.
10	Security	Protects network communication from unauthorized access, data modification, and other security threats.

Design issues for layers are the important functions and problems that network layers must handle to ensure reliable, efficient, and secure communication between devices.

Unit 2 : Data Link Layer

Data Link Layer– services provided to the network layer, framing and addressing; design issues of the data link layer– error control, flow control and reliable data transfer; error detection and correction techniques– parity, checksum, CRC and basic error correction concepts; data link layer protocols– elementary protocols and Stop-and-Wait protocol; sliding window protocols– pipelining, Go-Back-N and Selective Repeat protocols; example data link layer technologies– packet over SONET

Data Link Layer – Services Provided to the Network Layer, Framing and Addressing

The **Data Link Layer** is the second layer of the OSI model. It provides reliable communication between directly connected devices and is responsible for transferring data in the form of frames.

1. Services Provided to the Network Layer

The Data Link Layer provides the following services to the Network Layer:

1.1 Connectionless Unacknowledged Service

Data frames are sent without establishing a connection and without receiving any acknowledgement from the receiver.

Example:

Sender → Frame → Receiver

1.2 Connectionless Acknowledged Service

Data frames are sent without establishing a connection, but the receiver sends an acknowledgement after successfully receiving a frame.

Example:

Sender → Frame → Receiver

Sender ← ACK ← Receiver

1.3 Connection-Oriented Acknowledged Service

A connection is established before data transfer, frames are transmitted reliably, and the connection is terminated after communication is completed.

Example:

Connection Establishment → Data Transfer → Connection Termination

2. Framing

Framing is the process of dividing a continuous stream of bits received from the Physical Layer into manageable units called **frames**.

A frame generally consists of three main parts:

Header | Data | Trailer

- **Header:** Contains control information such as source and destination addresses.
- **Data:** Contains the packet received from the Network Layer.
- **Trailer:** Contains error-detection information such as FCS or CRC.

Framing helps the receiver identify the **beginning and end of each data unit**.

3. Addressing

The Data Link Layer uses **MAC (Media Access Control) addresses** to identify devices or network interfaces on a local network.

Example of MAC Address:

00:1A:2B:3C:4D:5E

A frame generally contains:

Destination MAC Address | Source MAC Address | Data

- **Source MAC Address:** Identifies the sender's network interface.
- **Destination MAC Address:** Identifies the receiver's network interface.

MAC Address vs IP Address

MAC Address: Used by the Data Link Layer for communication on the local network.

IP Address: Used by the Network Layer for logical addressing and routing between different networks.

Design Issues of the Data Link Layer – Error Control, Flow Control and Reliable Data Transfer

The **Data Link Layer** is responsible for providing reliable and efficient communication between two directly connected devices. The major design issues include **Error Control, Flow Control, and Reliable Data Transfer**.

1. Error Control

Error Control is the process of detecting and correcting errors that may occur when data is transmitted from the sender to the receiver.

Errors can occur due to **noise, interference, or signal distortion** during transmission.

The Data Link Layer uses techniques such as **CRC, checksum, acknowledgement, and retransmission** to handle errors.

Example:

Sender → Data → Receiver

Receiver detects error → Requests retransmission → Sender sends data again

2. Flow Control

Flow Control is a mechanism that controls the rate at which the sender transmits data so that a fast sender does not overwhelm a slow receiver.

If the sender sends data faster than the receiver can process it, the receiver's buffer may become full and data may be lost.

Example:

Fast Sender → → → → Slow Receiver

Flow control ensures that the sender adjusts its transmission rate according to the receiver's capacity.

Common flow control techniques include **Stop-and-Wait** and **Sliding Window**.

3. Reliable Data Transfer

Reliable Data Transfer ensures that data sent by the sender reaches the receiver **correctly, completely, and in the proper order**.

It uses mechanisms such as:

- **Acknowledgements (ACK):** Confirm successful reception of data.
- **Sequence Numbers:** Identify and arrange data frames in the correct order.
- **Timeouts:** Detect frames that may have been lost.
- **Retransmission:** Sends the lost or damaged frame again.

Example:

Sender → Frame 1 → Receiver

Sender ← ACK 1 ← Receiver

If Frame 1 is lost:

Sender → Frame 1 → ✕

Timeout → Sender retransmits Frame 1

Error Detection and Correction Techniques – Parity, Checksum, CRC and Basic Error Correction Concepts

During data transmission, the data may be changed or corrupted because of **noise, interference, or signal distortion**. Error detection and correction techniques are used to identify and correct these errors.

Error Detection: Determines whether an error has occurred in the received data.

Error

Correction: Identifies the error and recovers the correct original data.

1. Parity : Parity is a simple error detection technique in which an extra bit, called a **parity bit**, is added to the data to make the number of 1s either even or odd.

There are two types of parity:

Even Parity : The parity bit is added so that the total number of 1s becomes **even**.

Example:

Data = 1011001

Number of 1s = 4, which is already even.

Therefore:

Data + Parity Bit = 10110010

Odd Parity

The parity bit is added so that the total number of 1s becomes **odd**.

Example:

Data = 1011001

Number of 1s = 4, so the parity bit is 1.

Therefore:

Data + Parity Bit = 10110011

Limitation: Parity can detect many single-bit errors, but it cannot reliably detect all multiple-bit errors and cannot correct the error by itself.

2. Checksum

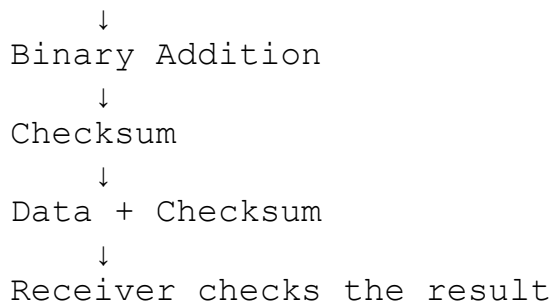
Checksum is an error detection technique in which the data is divided into fixed-size blocks, the blocks are added together, and the resulting value is used as a checksum.

Basic Process

1. Divide the data into fixed-size blocks.
2. Add all the blocks using binary addition.
3. Take the complement of the result to obtain the checksum.
4. Send the data along with the checksum.
5. The receiver performs the same calculation to check for errors.

Example:

Data Blocks



Use: Checksum is commonly used in networking protocols to detect errors in transmitted data.

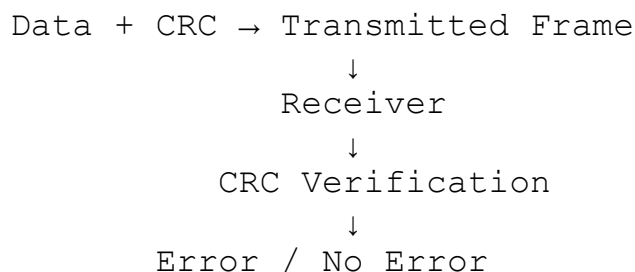
3. CRC (Cyclic Redundancy Check)

CRC is a powerful error detection technique that uses **binary polynomial division** to detect errors in a data frame.

Basic Process

1. The sender selects a **generator polynomial**.
2. Extra zeros are appended to the data.
3. Binary division is performed using the generator.
4. The remainder obtained is called the **CRC**.
5. The CRC is attached to the original data and transmitted.
6. The receiver performs the same division to check for errors.

Basic Representation:



Advantage: CRC is more powerful and reliable than simple parity for detecting common transmission errors.

4. Basic Error Correction Concepts

Error correction is the process of identifying and correcting errors in received data so that the original data can be recovered.

There are two basic approaches:

A. Forward Error Correction (FEC)

The sender adds extra redundant information to the data so that the receiver can detect and correct certain errors **without requesting retransmission**.

Example: Hamming Code.

```
Original Data
    ↓
Add Redundant Bits
    ↓
Transmit
    ↓
Receiver
    ↓
Detect and Correct Error
```

B. (Backward error correction)Retransmission-Based Error Correction

When an error is detected, the receiver requests the sender to **transmit the damaged or lost data again**.

This approach commonly uses:

- Acknowledgement (ACK)
- Negative acknowledgement (NAK)
- Timeout
- Retransmission

```
Sender → Data → Receiver
Sender ← NAK ← Receiver
Sender → Data Again → Receiver
```

Comparison of Techniques

Technique	Main Purpose	Basic Idea
Parity	Error Detection	Adds a parity bit to the data
Checksum	Error Detection	Calculates a value from data blocks
CRC	Error Detection	Uses polynomial division to generate a remainder
Hamming Code	Error Correction	Adds redundant bits to detect and correct errors
Retransmission	Error Correction	Requests damaged or lost data again

Data Link Layer Protocols – Elementary Protocols and Stop-and-Wait Protocol

The **Data Link Layer protocols** define rules for communication between two directly connected devices. They specify how frames are transmitted, received, acknowledged, and controlled during data communication.

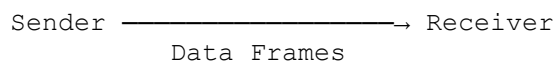
1. Elementary Data Link Layer Protocols

Elementary protocols are basic protocols used to understand the fundamental operations of data communication at the Data Link Layer.

1.1 Unrestricted Simplex Protocol

In this protocol, data is transmitted in **only one direction**, and the receiver is assumed to always be ready to receive data.

Example:



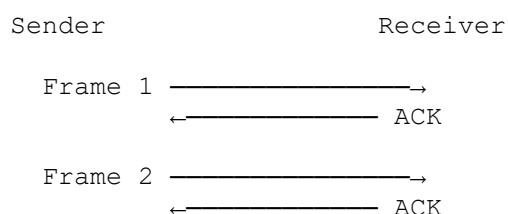
Characteristics:

- Communication is one-way.
 - No flow control is used.
 - No error control is provided.
 - The receiver is assumed to process data fast enough.
-

1.2 Simplex Stop-and-Wait Protocol

In this protocol, the sender sends **one frame at a time** and waits for an acknowledgement before sending the next frame.

Example:



This prevents the sender from sending too many frames before the receiver can process them.

2. Stop-and-Wait Protocol

The **Stop-and-Wait protocol** is a flow-control technique in which the sender transmits **one frame and waits for an acknowledgement (ACK)** before transmitting the next frame.

Working of Stop-and-Wait Protocol

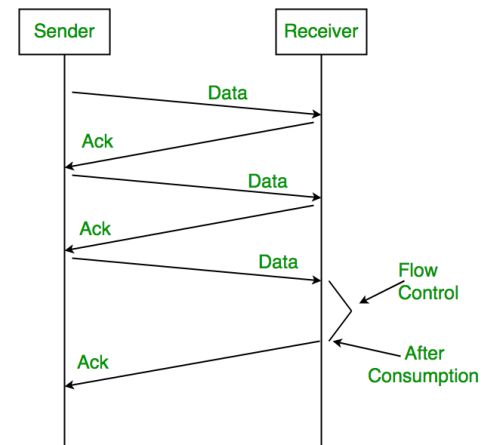
1. The sender sends one data frame.
2. The sender stops and waits for an acknowledgement.
3. The receiver receives the frame and sends an ACK.
4. After receiving the ACK, the sender sends the next frame.
5. This process continues until all data is transmitted.

Advantages

- Simple and easy to implement.
- Provides basic flow control.
- Prevents the receiver from being overloaded.
- Easy to understand and manage.

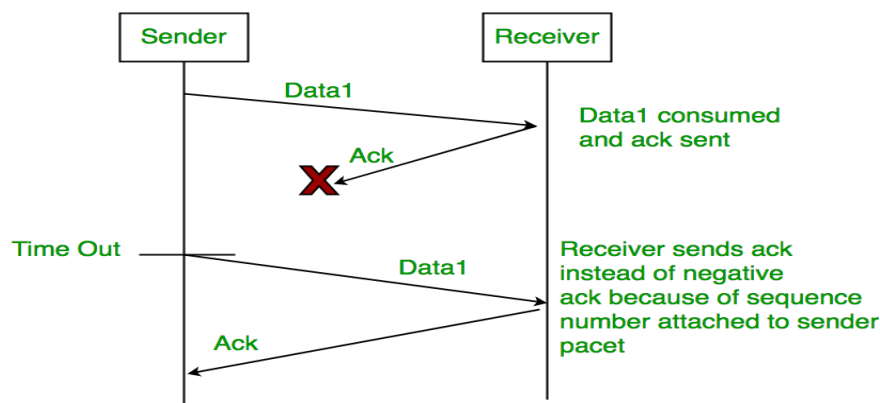
Disadvantages

- Slow because the sender waits for an ACK after every frame.
- Poor efficiency when the communication link has a long delay.
- Only one frame can be outstanding at a time.



3. Stop-and-Wait ARQ

Stop-and-Wait ARQ is a reliable data link layer protocol in which the sender transmits one frame at a time and waits for an acknowledgment (ACK) before sending the next frame. If the ACK is not received within a specified timeout period, the same frame is retransmitted to ensure reliability.



The sender uses a **timer** after sending a frame. If an ACK does not arrive within the expected time, the sender assumes that the frame or ACK was lost and retransmits the frame.

Sequence Numbers

Sequence numbers are used to identify duplicate frames.

For example:

Frame 0 → Frame 1 → Frame 0 → Frame 1

The receiver can determine whether a received frame is new or a duplicate.

Sliding Window Protocols – Pipelining, Go-Back-N, Selective Repeat and Packet over SONET

Sliding Window Protocol is a Data Link Layer protocol that allows the sender to transmit multiple frames before receiving acknowledgements, improving communication efficiency.

1. Pipelining

Pipelining is the process of sending multiple frames continuously without waiting for an acknowledgement after every frame.

Example:

Sender → Frame 1 → Frame 2 → Frame 3 → Frame 4 → Receiver

This improves **link utilization and transmission speed**, especially in networks with long propagation delays.

2. Go-Back-N Protocol

Go-Back-N ARQ is a sliding window protocol in which the sender can transmit multiple frames, but if one frame is lost or damaged, that frame and all subsequent frames are retransmitted.

Example:

Sender → F1 → F2 → F3 → F4 → Receiver

F3 × Lost

Sender → F3 → F4 → Receiver

Advantages: Simple and efficient compared with Stop-and-Wait.

Disadvantage: Correctly received frames after the damaged frame may need to be retransmitted, causing unnecessary retransmissions.

3. Selective Repeat Protocol

Selective Repeat ARQ is a sliding window protocol in which only the lost or damaged frames are retransmitted, while correctly received frames are kept by the receiver.

Example:

Sender → F1 → F2 → F3 → F4 → Receiver

F3 × Lost

Sender → F3 → Receiver

Advantages: More efficient because only erroneous or lost frames are retransmitted.

Disadvantage: More complex because the receiver must buffer out-of-order frames and maintain additional information.

4. Go-Back-N vs Selective Repeat

Go-Back-N

Selective Repeat

Retransmits the lost/error frame and all subsequent frames Retransmits only the lost/error frame

Less complex

More complex

Less efficient when errors are frequent

More efficient when errors are frequent

Receiver usually discards out-of-order frames

Receiver can buffer out-of-order frames

5. Packet over SONET

SONET (Synchronous Optical Networking) is a standard for transmitting digital data over optical fiber at high speeds. **Packet over SONET (PoS)** is a technology that carries IP packets directly over SONET links.

Basic Structure:

IP Packet → PPP → SONET → Optical Fiber → SONET → PPP → IP Packet

Features:

- Uses optical fiber for high-speed transmission.
- Provides high reliability and efficient data transfer.
- Commonly used in high-speed telecommunications and backbone networks.
- PPP is commonly used to encapsulate packets before transmission over SONET.

In short:

Pipelining → Sends multiple frames without waiting.

Go-Back-N → Retransmits the error frame and following frames.

Selective Repeat → Retransmits only the error frame.

Packet over SONET → Carries IP packets over high-speed SONET optical networks.